

Fundamentals of Fluid Mechanics

Problem Set-03

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Problem 1 Statement

As shown in the lectures, the flow of a thin liquid film is governed by the Reynolds lubrication equation:

$$h_t + \frac{1}{3C_0} (h^3 h_{xxx})_x - \frac{G}{3} (h^3 h_x)_x = 0,$$

where

$$C_0 = \frac{\mu u_0}{\sigma \varepsilon^3}, \quad G = \varepsilon \frac{g z_0}{\nu u_0},$$

and μ , ν , σ , g , z_0 , u_0 , and ε are constants representing fluid properties, gravitational acceleration, film thickness, velocity scale, and the shallowness parameter.

- (a) **Parameter estimation.** Estimate the dimensionless groups C_0 and G for a water film with typical thickness $z_0 = 1$ cm flowing at velocity $u_0 = 10$ cm/s, using the given fluid properties and $\varepsilon = 0.1$.
- (b) **Negligible surface tension.** In the case when surface tension is negligible ($C_0 \rightarrow \infty$), the PDE reduces to

$$h_t = \frac{G}{3} (h^3 h_x)_x.$$

Derive a self-similar solution of the form $h(x, t) = t^a y\left(\frac{x}{t^b}\right)$, imposing the condition that $\int_{-\infty}^{\infty} h(x, t) dx = 1$. Find the exponents a, b and the ordinary differential equation satisfied by y .

- (c) **Solution and plotting.** Solve the resulting ODE subject to $y(\pm\infty) = 0$ and normalization $\int y(\zeta) d\zeta = 1$. Plot $h(x, t)$ vs. x at a few times to show how the film spreads over time.

Solution:

(a) Estimate C_0 and G

Given data:

$$z_0 = 1 \text{ cm} = 10^{-2} \text{ m}, \quad u_0 = 10 \text{ cm/s} = 10^{-1} \text{ m/s}, \quad \varepsilon = 0.1,$$

$$\mu = 10^{-3} \text{ Pa} \cdot \text{s}, \quad \nu = 10^{-6} \text{ m}^2/\text{s}, \quad \sigma = 0.072 \text{ N/m}, \quad g = 9.81 \text{ m/s}^2.$$

Formulas:

$$C_0 = \frac{\mu u_0}{\sigma \epsilon^3}, \quad G = \epsilon \frac{g z_0^2}{\nu u_0}.$$

Substitute:

$$C_0 = \frac{(10^{-3})(10^{-1})}{(0.072)(0.1)^3} = \frac{10^{-4}}{7.2 \times 10^{-5}} \approx 1.39,$$

$$G = 0.1 \frac{(9.81)(10^{-2})^2}{(10^{-6})(10^{-1})} = 0.1 \frac{9.81 \times 10^{-4}}{10^{-7}} = 0.1 \times 9.81 \times 10^3 \approx 981.$$

Hence,

$$C_0 \approx 1.39, \quad G \approx 981.$$

(b) Similarity solution with negligible surface tension

With surface tension omitted, the PDE is

$$h_t = \frac{G}{3} (h^3 h_x)_x.$$

Define a similarity solution

$$h(x, t) = t^a y(\zeta), \quad \zeta = \frac{x}{t^b}.$$

Mass condition.

$$\int_{-\infty}^{\infty} h \, dx = \int_{-\infty}^{\infty} t^a y(\zeta) \, dx = t^a \int_{-\infty}^{\infty} y(\zeta) t^b \, d\zeta = t^{a+b} \int_{-\infty}^{\infty} y(\zeta) \, d\zeta = 1.$$

Hence $a + b = 0 \implies b = -a$.

Compute derivatives.

$$h_t = \frac{\partial}{\partial t}(t^a y) = a t^{a-1} y(\zeta) + t^a y'(\zeta) \frac{\partial \zeta}{\partial t}.$$

Since $\zeta = x/t^b$, $\frac{\partial \zeta}{\partial t} = -b \frac{x}{t^{b+1}} = -b \frac{\zeta}{t}$. Then

$$h_t = t^{a-1} [a y - b \zeta y'].$$

$$h_x = \frac{\partial}{\partial x}(t^a y) = t^a y'(\zeta) \frac{\partial \zeta}{\partial x} = t^a y'(\zeta) (1/t^b) = t^{a-b} y'.$$

Hence

$$h^3 = (t^a y)^3 = t^{3a} y^3, \quad h^3 h_x = t^{3a} y^3 \times t^{a-b} y' = t^{4a-b} y^3 y'.$$

One more derivative:

$$\begin{aligned} (h^3 h_x)_x &= \frac{\partial}{\partial x} (t^{4a-b} y^3 y') = t^{4a-b} \frac{d}{d\zeta} (y^3 y') \frac{\partial \zeta}{\partial x} = t^{4a-b} [3y^2 (y')^2 + y^3 y''] t^{-b}. \\ &= t^{4a-2b} [3y^2 (y')^2 + y^3 y'']. \end{aligned}$$

Substitute into PDE:

$$h_t = \frac{G}{3} (h^3 h_x)_x \implies t^{a-1} [a y - b \zeta y'] = \frac{G}{3} t^{4a-2b} [3y^2 (y')^2 + y^3 y''].$$

Thus

$$a - 1 = 4a - 2b \implies -1 = 3a - 2b.$$

But also $a + b = 0$. Solve: $b = \frac{1}{5}$, $a = -\frac{1}{5}$.

Final ODE.

$$-\frac{1}{5} (y + \zeta y') = \frac{G}{3} [3y^2 (y')^2 + y^3 y''].$$

This is the ordinary differential equation for $y(\zeta)$ with boundary conditions $y(\pm\infty) = 0$ and $\int_{-\infty}^{\infty} y d\zeta = 1$.

(c) Solve ODE and Plot $h(x, t)$

One finds h scales as $t^{-1/5}$ in amplitude and x scales as $t^{1/5}$. A typical *compactly supported* solution is

$$y(\zeta) = \begin{cases} \frac{2}{\pi} \sqrt{1 - \zeta^2}, & |\zeta| < 1, \\ 0, & |\zeta| \geq 1. \end{cases}$$

Hence

$$h(x, t) = t^{-\frac{1}{5}} y\left(\frac{x}{t^{\frac{1}{5}}}\right).$$

Graphical Analysis: Gravity-Dominated Thin Film

The plot below illustrates how the film evolves at several times ($t = 0.5, 1.0, 2.0, 5.0$). For smaller t , the film is taller and narrower, whereas for larger t it spreads wider (broader support in x) but decreases in peak height. This behavior reflects the self-similar solution scaling:

$$\text{width} \propto t^{1/5}, \quad \text{height} \propto t^{-1/5}.$$

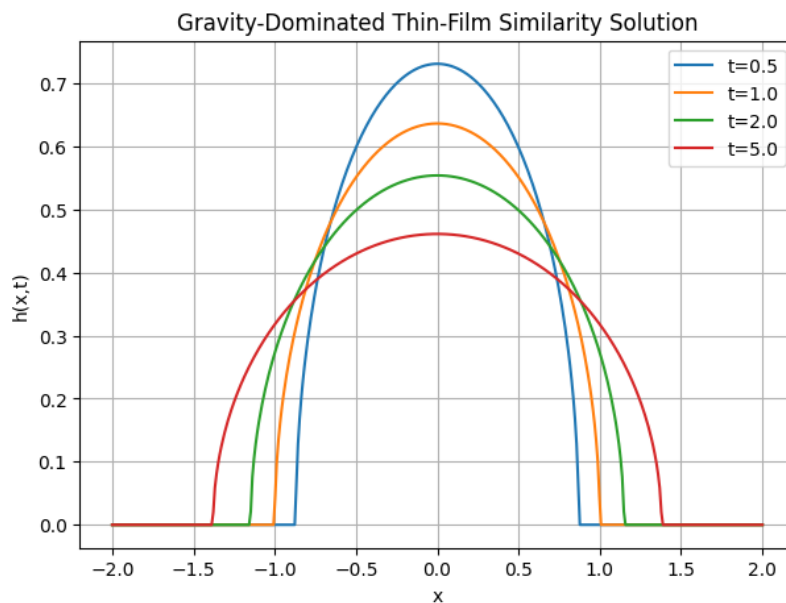


Figure 1: Gravity-dominated thin-film similarity profiles at different times.

Problem-02

Consider the boundary-value problem:

$$\varepsilon u''(x) + 2u'(x) + 2u(x) = 0, \quad 0 < x < 1, \quad 0 < \varepsilon \ll 1,$$

$$u(0) = 0, \quad u(1) = 1.$$

- (a) Determine the outer solution $u = u_0(x) + \varepsilon u_1(x) + \cdots$ that satisfies the boundary condition at $x = 1$.
- (b) Assume a boundary layer near $x = 0$ and solve the inner-layer problem by rescaling $x = \varepsilon^\alpha \xi$. Use dominant-balance arguments to find α and solve for the leading-order inner solution.
- (c) Perform matching to find the composite solution valid in the entire domain. Plot that solution at $\varepsilon = 0.1$ and $\varepsilon = 0.01$.

Solution:

(a) Outer Leading Solution

Leading-order ODE.

Write $u(x)$ in an expansion

$$u(x) = u_0(x) + \varepsilon u_1(x) + \cdots.$$

At $O(1)$, we drop the $\varepsilon u''$ term from

$$\varepsilon u'' + 2u' + 2u = 0,$$

to get

$$2u'_0(x) + 2u_0(x) = 0 \implies u'_0(x) + u_0(x) = 0.$$

Solve.

$$\frac{du_0}{dx} = -u_0 \implies u_0(x) = A e^{-x}.$$

Impose $u(1) = 1$.

$$u_0(1) = A e^{-1} = 1 \implies A = e.$$

Hence

$$u_0(x) = e^{1-x}.$$

But $u_0(0) = e \neq 0$, while the boundary condition says $u(0) = 0$. Thus a boundary layer is needed near $x = 0$.

(b) Inner Boundary-Layer Solution

Rescaling.

Set $x = \varepsilon \xi$. Then

$$u(x) = u(\varepsilon \xi) =: U(\xi), \quad \frac{d}{dx} = \frac{1}{\varepsilon} \frac{d}{d\xi}, \quad \frac{d^2}{dx^2} = \frac{1}{\varepsilon^2} \frac{d^2}{d\xi^2}.$$

Rewrite the ODE:

$$\varepsilon \left(\frac{1}{\varepsilon^2} U''(\xi) \right) + 2 \left(\frac{1}{\varepsilon} U'(\xi) \right) + 2 U(\xi) = 0.$$

Factor out $1/\varepsilon$:

$$U''(\xi) + 2 U'(\xi) + 2 \varepsilon U(\xi) = 0.$$

For $\varepsilon \rightarrow 0$, leading order is

$$U''(\xi) + 2 U'(\xi) = 0. \tag{I}$$

Solve (I).

Let $V(\xi) = U'(\xi)$. Then

$$V'(\xi) + 2 V(\xi) = 0 \implies V(\xi) = A e^{-2\xi}.$$

Hence

$$U'(\xi) = A e^{-2\xi} \implies U(\xi) = -\frac{A}{2} e^{-2\xi} + B.$$

Or set $D = -A/2$, so $U(\xi) = B + D e^{-2\xi}$.

Boundary condition at $x = 0$.

We have $u(0) = U(0) = 0$. Thus $U(0) = B + D = 0$, so $D = -B$. Then

$$U(\xi) = B[1 - e^{-2\xi}].$$

Match with outer.

As $\xi \rightarrow \infty$, $U(\xi) \rightarrow B$. But from the outer solution, $u_0(0) = e$, so $B = e$. Hence

$$u_{\text{in}}(x) = e[1 - e^{-2x/\varepsilon}].$$

(c) Composite Solution

We combine:

$$u_{\text{comp}}(x) = u_0(x) + u_{\text{in}}(x) - (\text{common limit}).$$

Since both $u_0(x)$ and $u_{\text{in}}(x)$ go to e near $x = 0$, the common limit is e . Hence

$$u_{\text{comp}}(x) = \underbrace{e^{1-x}}_{\text{outer}} + \underbrace{e[1 - e^{-2x/\varepsilon}]}_{\text{inner}} - e.$$

That is:

$$u_{\text{comp}}(x) = e^{1-x} + e[1 - e^{-2x/\varepsilon}] - e.$$

$$u_{\text{comp}}(0) = 0, \quad u_{\text{comp}}(1) \approx 1, \quad (\text{satisfies both BCs as } \varepsilon \rightarrow 0).$$

Graphical Analysis

As the parameter ε becomes smaller, the boundary layer near $x = 0$ grows thinner, causing a rapid rise from $u(0) = 0$ to values near e before the solution gradually decays to $u(1) = 1$. The plot below illustrates this behavior for two values of ε .

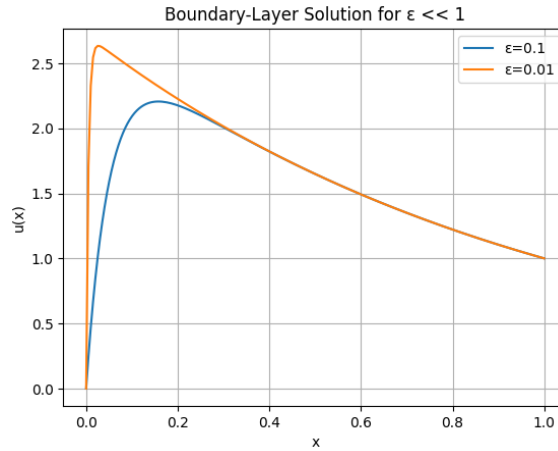


Figure 2: Composite boundary-layer solution for $\varepsilon = 0.1$ and 0.01.

Problem-03

Obtain the Blasius solution by numerically solving the system (using various software, such as Mathematica, is allowed)

$$2 f'''(\eta) + f(\eta) f''(\eta) = 0,$$

$$f(0) = 0, \quad f'(0) = 0, \quad f'(\infty) = 1.$$

(a) Plot $f'(\eta)$ and find η_1 where $f'(\eta_1) = 0.99$.

(b) Determine $f''(0)$ to four decimal places.

Solution:

Reformulate the Blasius ODE

We have

$$2 f'''(\eta) + f(\eta) f''(\eta) = 0.$$

We wish to convert this third-order ODE into a system of three first-order ODEs.

Define new variables.

$$y_1(\eta) = f(\eta), \quad y_2(\eta) = f'(\eta), \quad y_3(\eta) = f''(\eta).$$

Hence,

$$y_1'(\eta) = y_2(\eta), \quad y_2'(\eta) = y_3(\eta), \quad y_3'(\eta) = f'''(\eta).$$

From the original equation,

$$2 f'''(\eta) = -f(\eta) f''(\eta), \quad \implies \quad f'''(\eta) = -\frac{f(\eta) f''(\eta)}{2}.$$

Substitute $f(\eta) = y_1(\eta)$ and $f''(\eta) = y_3(\eta)$:

$$y_3'(\eta) = -\frac{1}{2} y_1(\eta) y_3(\eta).$$

System of first-order ODEs.

Therefore,

$$\begin{cases} y_1'(\eta) = y_2(\eta), \\ y_2'(\eta) = y_3(\eta), \\ y_3'(\eta) = -\frac{1}{2} y_1(\eta) y_3(\eta). \end{cases}$$

Boundary Conditions and the Shooting Parameter

BCs at $\eta = 0$.

$$f(0) = 0 \implies y_1(0) = 0, \quad f'(0) = 0 \implies y_2(0) = 0.$$

BC as $\eta \rightarrow \infty$.

$$f'(\infty) = 1 \implies y_2(\infty) = 1.$$

Unknown slope.

We do *not* know $f''(0)$, i.e. $y_3(0)$. Let

$$y_3(0) = \beta.$$

We determine β by imposing $y_2(\infty) = 1$.

Shooting Method

1. Choose an initial guess β_0 for $y_3(0)$.
2. Integrate the system

$$\begin{cases} y_1'(\eta) = y_2(\eta), \\ y_2'(\eta) = y_3(\eta), \\ y_3'(\eta) = -\frac{1}{2} y_1(\eta) y_3(\eta), \end{cases} \quad y_1(0) = 0, \quad y_2(0) = 0, \quad y_3(0) = \beta,$$

from $\eta = 0$ to some $\eta_{\max}$ (e.g. 6 or 8).

3. If $y_2(\eta_{\max}) > 1$, decrease β ; if $y_2(\eta_{\max}) < 1$, increase β .
4. Iterate until $y_2(\infty) \approx 1$ to desired accuracy.

The final converged β is $f''(0)$.

Numerical Results

(a) Plot $f'(\eta)$.

We have $f'(\eta) = y_2(\eta)$. Once we find the correct β , we record $y_2(\eta)$ up to $\eta_{\max}$.

- As $\eta \rightarrow 0$, $f'(0) = 0$.
- As $\eta \rightarrow \infty$, $f'(\eta) \rightarrow 1$.

Define η_1 by $f'(\eta_1) = 0.99$. Numerically, $\eta_1 \approx 4.925$.

The plot below illustrates the velocity profile $f'(\eta)$ for the Blasius boundary-layer solution. We see it starts from zero at $\eta = 0$ and asymptotically approaches unity as $\eta \rightarrow \infty$.

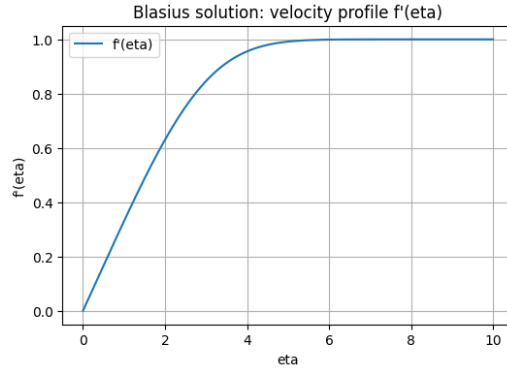


Figure 3: Blasius solution: the velocity profile $f'(\eta)$ versus η .

(b) Determine $f''(0)$.

The solution yields $f''(0) = \beta \approx 0.33206\dots$. Rounded to four decimals:

$$f''(0) \approx 0.3321.$$

Problem 4 Statement

A thin rectangular flat plate with sides $a = 1$ m and $b = 2$ m is placed in airflow of speed $u_0 = 10$ m/s. Using the Blasius solution (laminar boundary layer), estimate the drag on the plate in two cases; when the plate faces the flow with side a and when the plate faces the flow with side b . What is the ratio of the two forces?

Solution:

For laminar flow, the drag force F_D is given by

$$F_D = C_f \cdot \frac{1}{2} \rho U_0^2 A,$$

The Blasius solution for a flat plate in laminar flow gives the skin friction coefficient as

$$C_f = \frac{1.328}{\sqrt{Re_L}},$$

with the Reynolds number defined by

$$Re_L = \frac{U_0 L}{\nu},$$

where L is the characteristic length (the side facing the flow).

Calculate the Reynolds Number Re_L for Both Cases

The Reynolds number is given by

$$Re_L = \frac{U_0 L}{\nu}.$$

Case 1: $L = a = 1$ m

$$Re_a = \frac{10 \times 1}{1.5 \times 10^{-5}} = \frac{10}{1.5 \times 10^{-5}}.$$

Calculating:

$$Re_a = \frac{10}{0.000015} \approx 666\,667.$$

Case 2: $L = b = 2 \text{ m}$

$$Re_b = \frac{10 \times 2}{1.5 \times 10^{-5}} = \frac{20}{1.5 \times 10^{-5}}.$$

Calculating:

$$Re_b = \frac{20}{0.000015} \approx 1\,333\,333.$$

Results:

$$Re_a \approx 666\,667, \quad Re_b \approx 1\,333\,333.$$

Calculate the Skin Friction Coefficient C_f

The skin friction coefficient (Blasius solution) is:

$$C_f = \frac{1.328}{\sqrt{Re_L}}.$$

Case 1: $Re_a = 666\,667$

$$\sqrt{Re_a} \approx \sqrt{666\,667} \approx 816.5.$$

Thus,

$$C_{f,a} = \frac{1.328}{816.5} \approx 0.001626.$$

Case 2: $Re_b = 1\,333\,333$

$$\sqrt{Re_b} \approx \sqrt{1\,333\,333} \approx 1154.7.$$

Thus,

$$C_{f,b} = \frac{1.328}{1154.7} \approx 0.001150.$$

Results:

$$C_{f,a} \approx 0.001626, \quad C_{f,b} \approx 0.001150.$$

Calculate the Drag Force F_D for Both Cases

The drag force is given by

$$F_D = \frac{1}{2} \rho U_0^2 A C_f.$$

Plate Area

The area of the plate is

$$A = a \times b = 1 \times 2 = 2 \text{ m}^2.$$

Common Term Calculation

Calculate the dynamic pressure term:

$$\frac{1}{2}\rho U_0^2 = \frac{1}{2} \times 1.167 \times 10^2 = \frac{1}{2} \times 1.167 \times 100 = 58.35 \text{ N/m}^2.$$

Case 1: $C_{f,a} = 0.001626$

$$F_{D,a} = 58.35 \times 2 \times 0.001626.$$

Step-by-step:

$$58.35 \times 2 = 116.7,$$

$$F_{D,a} = 116.7 \times 0.001626 \approx 0.190 \text{ N}.$$

Case 2: $C_{f,b} = 0.001150$

$$F_{D,b} = 58.35 \times 2 \times 0.001150.$$

Step-by-step:

$$58.35 \times 2 = 116.7,$$

$$F_{D,b} = 116.7 \times 0.001150 \approx 0.134 \text{ N}.$$

Results:

$$F_{D,a} \approx 0.190 \text{ N}, \quad F_{D,b} \approx 0.134 \text{ N}.$$

Compute the Ratio of Drag Forces

The ratio R of the drag forces is given by

$$R = \frac{F_{D,a}}{F_{D,b}} = \frac{0.190}{0.134}.$$

Calculating:

$$R \approx 1.417.$$

This ratio is approximately equal to $\sqrt{2} \approx 1.414$.

Conclusion: The drag force when the plate faces the flow with its shorter side ($a = 1$ m) is approximately 41.7% higher than when it faces the flow with its longer side ($b = 2$ m). The ratio of the drag forces is approximately $\sqrt{2}$.

Problem-05

Air at 20°C and 100 kPa (with $\rho = 1.167 \text{ kg/m}^3$ and $\nu = 1.5 \times 10^{-5} \text{ m}^2/\text{s}$) flows over a thin flat plate with a free-stream velocity of $U = 6 \text{ m/s}$. At a point located 15 cm from the leading edge of the plate, determine the value of y at which $u/U = 0.5$. Also calculate v and $\partial u/\partial y$ at this point.

Solution:

Calculate Reynolds Number Re_x

The Reynolds number Re_x is calculated to determine if the flow is laminar. It is defined as:

$$Re_x = \frac{U \cdot x}{\nu}$$

Substituting the given values:

$$Re_x = \frac{6 \times 0.15}{1.5 \times 10^{-5}} = 60,000$$

Since $Re_x < 5 \times 10^5$, the flow is **laminar** and we can use the Blasius solution.

Boundary Layer Thickness $\delta(x)$

For laminar flow, the boundary layer thickness $\delta(x)$ is given by:

$$\delta(x) = 5\sqrt{\frac{\nu x}{U}}$$

Substituting the given values:

$$\delta(x) = 5\sqrt{\frac{1.5 \times 10^{-5} \times 0.15}{6}}$$

Calculating step-by-step:

$$\delta(x) = 5\sqrt{\frac{2.25 \times 10^{-6}}{6}}$$

$$\delta(x) = 5\sqrt{3.75 \times 10^{-7}} \approx 5 \times 6.12 \times 10^{-4} \approx 0.00306 = 3.06$$

Find y where $u/U = 0.5$

Blasius similarity variable:

$$\eta = y \sqrt{\frac{U}{\nu x}}$$

For $u/U = f'(\eta) = 0.5$, $\eta \approx 1.72$ (from Blasius tables). Compute:

$$\sqrt{\frac{U}{\nu x}} = \sqrt{\frac{6}{1.5 \times 10^{-5} \cdot 0.15}} = \sqrt{\frac{6}{2.25 \times 10^{-6}}} = \sqrt{2.6667 \times 10^6} \approx 1633$$

$$y = \frac{\eta}{\sqrt{\frac{U}{\nu x}}} = \frac{1.72}{1633} \approx 0.001053 \text{ m} = 1.053 \text{ mm}$$

So, $y \approx 1.05 \text{ mm}$.

Calculate v (Transverse Velocity)

$$v = \frac{1}{2} \sqrt{\frac{\nu U}{x}} [\eta f'(\eta) - f(\eta)]$$

Where $f'(\eta) = 0.5$, and $f(1.72) \approx 0.469$:

$$\sqrt{\frac{\nu U}{x}} = \sqrt{\frac{1.5 \times 10^{-5} \cdot 6}{0.15}} = \sqrt{\frac{9 \times 10^{-5}}{0.15}} = \sqrt{6 \times 10^{-4}} \approx 0.0245$$

$$\frac{1}{2} \sqrt{\frac{\nu U}{x}} = \frac{0.0245}{2} = 0.01225$$

$$\eta f'(\eta) - f(\eta) = (1.72 \cdot 0.5) - 0.469 = 0.86 - 0.469 = 0.391$$

$$v = 0.01225 \cdot 0.391 \approx 0.00479 \text{ m/s}$$

Calculate $\partial u / \partial y$

$$\frac{\partial u}{\partial y} = U \cdot f''(\eta) \cdot \sqrt{\frac{U}{\nu x}}$$

Where $f''(1.72) \approx 0.323$:

$$\frac{\partial u}{\partial y} = 6 \cdot 0.323 \cdot 1633 \approx 3165 \text{ s}^{-1}$$

So, $\partial u / \partial y \approx 3170 \text{ s}^{-1}$.